

PART A

1) Total resistance in ohms

a) Total resistance

In series

$$R = r_1 + r_2 + r_3$$

$$20\Omega + 30\Omega + 50\Omega$$

$$= \underline{\underline{100\Omega}}$$

b) Total resistance in parallel

$$R = \left(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \right)^{-1}$$

$$\frac{1}{20} + \frac{1}{100} + \frac{1}{50} = \left(\frac{2}{25} \right)^{-1} = \frac{25}{2}$$

$$= \underline{\underline{12.5\Omega}}$$

2) Total current from the power supply

$$V = IR \quad \therefore \quad I = \frac{V}{R}$$

a) $V = 125V$

$R = 100\Omega$

$$I = \frac{V}{R} = \frac{125}{100} = \underline{\underline{1.25 A}}$$

b) $V = 125V$

$R = 12.5\Omega$

$$I = \frac{V}{R} = \frac{125}{12.5} = \underline{\underline{10 A}}$$

3) Current through each resistor

a) Current in series is the same in each resistor
Therefore 20Ω , 30Ω and 50Ω all have $= \underline{\underline{1.25 A}}$

b) Current in parallel

$$20\Omega = \frac{125}{20} = \underline{\underline{6.25 A}}$$

$$100\Omega = \frac{125}{100} = \underline{\underline{1.25 A}}$$

$$50\Omega = \frac{125}{50} = \underline{\underline{2.5 A}}$$

4) Voltage drop

a) Series

$$V = IR$$

$$20\Omega = 1.25 \times 20 = 25V = \underline{\underline{25V}}$$

$$30\Omega = 1.25 \times 30 = 37.5V = \underline{\underline{37.5V}}$$

$$50\Omega = 1.25 \times 50 = 62.5V = \underline{\underline{62.5V}}$$

b) Parallel

Voltage is the same across all branches

therefore

$$20\Omega = \underline{\underline{125V}}$$

$$100\Omega = \underline{\underline{125V}}$$

$$50\Omega = \underline{\underline{125V}}$$

$$= \underline{\underline{125V}}$$

5) Power through each resistor

$$P = VI \quad \text{or} \quad P = I^2 R$$

a) Series

$$I = 1.25$$

$$20\Omega = 25 \times 1.25 = 31.25W = \underline{\underline{31.25W}}$$

$$30\Omega = 37.5 \times 1.25 = 46.875W = \underline{\underline{46.875W}}$$

$$50\Omega = 62.5 \times 1.25 = 78.125W = \underline{\underline{78.125W}}$$

b) Parallel

$$P = VI$$

$$20\Omega = 125 \times 6.25 = 781.25 \text{ W} //$$

$$100\Omega = 125 \times 1.25 = 156.25 \text{ W} //$$

$$50\Omega = 125 \times 2.5 = 312.5 \text{ W} //$$